Our literature review revolved around three research papers advancing building height estimation. First Cao and Huang's M3Net [1], that leverages high-resolution ZY-3 multi-view imagery over 42 Chinese cities. Second, Wang and Co's MF-BHNet [2], that fuses Sentinel-1 SAR and Sentinel-2 optical data. Third, Sana Jabbar and Murtaza Taj's Stereollax Net [3], that applies a stereo parallax-based approach using RGB imagery.

Methodologically, M3Net employs a multi-spectral, multi-view, and multi-task framework to reduce uncertainty in building height predictions. MF-BHNet uses a hybrid multimodal encoder with a transformer-based cross-modal module to integrate complementary features from SAR and optical images. Stereollax Net combines analytical stereo parallax formulations with a multiscale differential shortcut connection to enhance performance.

Our approach aims to build on these advancements by incorporating vision transformers and hybrid deep learning models to enhance feature extraction and fusion. We will preprocess satellite data and train both baseline and advanced multimodal fusion models, integrating transformers with CNNs to improve existing benchmarks. By integrating vision transformers and other state-of-the-art deep learning strategies to further refine feature extraction and fusion, we hope to improve the accuracy and robustness of building height estimation.

[1] X. Cao and X. Huang, "A Deep Learning Method for Building Height Estimation Using High-Resolution Multi-View Imagery Over Urban Areas: A Case Study of 42 Chinese Cities," *ScienceDirect*, 2021.

[2] H. Wang, Y. Liu, and X. Zhang, "MF-BHNet: A Hybrid Multimodal Fusion Network for Building Height Estimation Using Sentinel-1 and Sentinel-2 Imagery," *Remote Sensing*, 2024.

[3] S. Jabbar and I. Taj, "Stereollax Net: Stereo Parallax-Based Deep Learning Network for Building Height Estimation," *SpringerLink*, 2024.